

SOLUTION OF NON-LINEAR EQUATIONS

(A.K.A. ROOT FINDING)

WILL CONSIDER TWO CASES

1. $f(x) = 0$ "1-DIMENSIONAL"
2. $\underline{f}(\underline{x}) = \underline{0}$ "0- " "

$$\underline{x} \equiv [x_1, x_2, \dots, x_d]$$

$$\underline{f} \equiv [f_1(x_1, x_2, \dots, x_d), \dots, f_d(x_1, x_2, \dots, x_d)]$$

1. SOLVING NONLINEAR EQNS IN ONE VAR

• TWO TECHNIQUES: BOTH PRE SUPPOSE INFORMATION ABOUT LOCATION OF ROOT

1. BISECTION (BINARY SEARCH)
2. NEWTON'S METHOD (NEWTON-RAPHSON METHOD)

1.1 PRELIMINARIES

WANT TO FIND ONE OR MORE ROOTS OF

$$f(x) = 0 \quad (1)$$

CAN ALWAYS CAST IN THIS CANONICAL FORM

• DEFINITION: GIVEN CANONICAL EQN $f(x) = 0$ THE RESIDUAL OF THE EQN FOR A GIVEN VALUE OF x IS JUST f EVALUATED AT THAT VALUE

• ITERATIVE TECHNIQUE: ASSUME $f(x) = 0$ HAS A ROOT AT $x = x^*$; THEN CONSIDER SEQUENCE OF ESTIMATES (ITERATES) OF x^* , $x^{(n)}$

$$x^{(0)} \rightarrow x^{(1)} \rightarrow x^{(2)} \rightarrow \dots \rightarrow x^{(n)} \rightarrow x^{(n+1)} \rightarrow \dots \rightarrow x^*$$

• ASSOCIATED WITH THESE ESTIMATES ARE THE RESIDUALS

$$\begin{array}{ccccccc} r^{(0)} & \rightarrow & r^{(1)} & \rightarrow & r^{(2)} & \rightarrow & \dots \rightarrow r^{(n)} \rightarrow r^{(n+1)} \rightarrow \dots \rightarrow 0 \\ \text{"} & & \text{"} & & \text{"} & & \text{"} \\ f(x^{(0)}) & & f(x^{(1)}) & & f(x^{(2)}) & & f(x^{(n)}) \quad f(x^{(n+1)}) \quad \dots \rightarrow f(x^*) \end{array}$$

LOCATING A ROOT $\equiv$ DRIVING THE RESIDUAL TO 0

- CONVERGENCE: ITERATIVE METHOD $\rightarrow$ WHEN DO WE STOP ITERATION?

- FOR ROOT FINDING CASE, IS NATURAL TO STOP WHEN

$$|\delta x^{(n)}| \equiv |x^{(n+1)} - x^{(n)}| \leq \epsilon \quad (2)$$

WHERE ϵ IS SOME PRESCRIBED CONVERGENCE CRITERION

- BETTER IDEA IS TO USE A "RELATIVIZED" δx

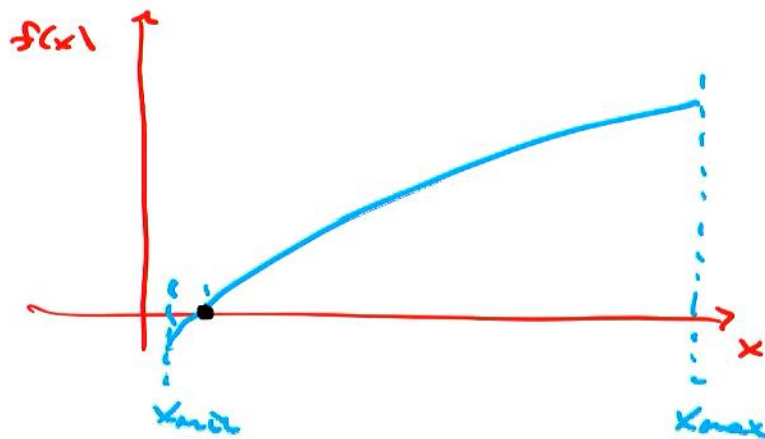
$$\frac{|\delta x^{(n)}|}{|x^{(n+1)}|} \leq \epsilon \quad (3)$$

BUT SHOULD SWITCH OVER TO "ABSOLUTE" FORM (2)

... $|x^{(n+1)}|$ BECOMES "TOO SMALL"

1.1 BISECTION

- ASSUME THAT WE KNOW SOMEHOW THAT THERE IS A UNIQUE ROOT OF $f(x) = 0$ IN INTERVAL $x_{\min} \leq x \leq x_{\max}$



- IN PARTICULAR, WILL DEMAND THAT

$$f(x_{\min}) \cdot f(x_{\max}) < 0$$

- BISECTION WORKS BY STARTING WITH THE INITIAL BRACKET $[x_{\min}, x_{\max}]$ HAVING WIDTH $\delta x_0 = x_{\max} - x_{\min}$ THEN GENERATING ADDITIONAL BRACKETS HAVING WIDTHS $\delta x_1 = \delta x_0 / 2$, $\delta x_2 = \delta x_0 / 4$, $\delta x_3 = \delta x_0 / 8$, ...

EACH OF WHICH CONTAINS THE ROOT

• ALGORITHM: PSEUDO CODE

converged = false

$f_{min} = f(x_{min})$

while not converged do

$x_{mid} = (x_{min} + x_{max}) / 2$

$f_{mid} = f(x_{mid})$

if $f_{mid} == 0$ then

converged = true

elseif $f_{mid} * f_{min} < 0$

$x_{max} = x_{mid}$

else

$x_{min} = x_{mid}$

$f_{min} = f_{mid}$

end if

if $(x_{max} - x_{min}) / \text{abs}(x_{mid}) \leq \text{epsilon}$ then

converged = true

end if

end while

root = x_{mid}

• AFTER THE 2^{th} BISECTION, THE WIDTH OF THE SEARCH INTERVAL IS

$$\delta x_2 = \delta x_1 2^{-2} = \frac{x_{max} - x_{min}}{2^2}$$

AND δx_2 PROVIDES AN UPPER BOUND ON THE ERROR IN THE ROOT ESTIMATE (ROOT CAN LIE ANYWHERE IN THE SMALLEST INTERVAL)